Marketing Campaign Performance Analysis

Tools Used: Python, MySQL, Power BI

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1. Introduction

Marketing campaigns generate large amounts of data related to impressions, clicks, conversions, revenue, and costs. Analyzing this data helps businesses understand which marketing channels and campaigns are most effective.

The objective of this project is to analyze marketing campaign data and build a dashboard that provides insights into campaign performance, channel effectiveness, and return on investment (ROI).

The project follows a complete data analytics workflow including data cleaning, data storage, SQL analysis, and visualization.

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2. Project Objectives

The main objectives of this project are:

- **Analyze overall marketing performance**
- **Identify the best performing marketing channels**
- **Discover top campaigns based on ROI**
- **Compare conversion rates across channels**
- **Analyze revenue trends over time**
- **Build an interactive dashboard for visualization**

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3. Dataset Overview

The dataset contains marketing campaign performance data with the following columns:

Column	Description
campaign_id	Unique identifier for each campaign
channel	Marketing channel (Email, Social Media, PPC, etc.)

Column	Description
impressions	Number of times the ad was shown
clicks	Number of clicks received
conversions	Number of successful conversions
revenue	Revenue generated from the campaign
cost	Marketing cost
date	Campaign date
CTR	Click Through Rate
conversion_rate	Conversion percentage
CPC	Cost per click
ROI	Return on Investment

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4. Data Cleaning (Python)

The dataset was first cleaned using Python and Pandas before loading it into MySQL.

The following steps were performed:

- **Removed missing or inconsistent values**
- **Ensured numeric columns were properly formatted**
- **Standardized date format**
- **Exported the cleaned dataset as a CSV file**

The cleaned dataset was saved as:

clean_marketing_campaign_data.csv

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5. Data Storage and Processing (MySQL)

The cleaned dataset was loaded into MySQL for further analysis.

Step 1: Create Table

```
CREATE TABLE marketing_campaigns (
campaign_id TEXT,
channel TEXT,
impressions INT,
clicks INT,
conversions INT,
revenue FLOAT,
cost FLOAT,
date DATE,
CTR FLOAT,
conversion_rate FLOAT,
CPC FLOAT,
ROI FLOAT
);
```

Step 2: Load Data into MySQL

```
LOAD DATA INFILE
'C:/ProgramData/MySQL/MySQL Server
8.0/Uploads/clean_marketing_campaign_data.csv'
INTO TABLE marketing_campaigns
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

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6. SQL Analysis

Several SQL queries were used to analyze marketing campaign performance.

Total Marketing Performance

This query calculates overall marketing metrics.

```
SELECT
SUM(impressions) AS Total_Impressions,
SUM(clicks) AS Total_clicks,
SUM(conversions) AS Total_conversions,
SUM(revenue) AS Total_Revenue,
SUM(cost) AS Total_cost
FROM marketing_campaigns;
```

Best Performing Marketing Channel

This query identifies which channel generates the highest revenue and ROI.

```
SELECT
channel,
ROUND(SUM(revenue),2) AS Total_revenue,
ROUND(SUM(cost),2) AS Total_Cost,
ROUND(SUM(ROI),2) AS Total_ROI
FROM marketing_campaigns
GROUP BY channel
ORDER BY Total_revenue DESC;
```

Top 10 Campaigns by ROI

This query identifies campaigns that generate the highest return on investment.

```
SELECT
campaign_id,
channel,
revenue,
cost,
roi
FROM marketing_campaigns
ORDER BY ROI DESC
LIMIT 10;
```

Channel Performance by Conversion Rate

This query analyzes which marketing channel converts users most effectively.

```
SELECT
channel,
AVG(conversion_rate) AS avg_conversion_rate
FROM marketing_campaigns
GROUP BY channel
ORDER BY avg_conversion_rate DESC;
```

CTR Performance by Channel

This query compares click-through rates across marketing channels.

```
SELECT
channel,
AVG(ctr) AS avg_ctr
FROM marketing_campaigns
GROUP BY channel
ORDER BY avg_ctr;
```

Revenue Trend Over Time

This query analyzes how revenue changes over time.

```
SELECT
date,
ROUND(SUM(revenue),2) AS daily_revenue
FROM marketing_campaigns
GROUP BY date
ORDER BY date;
```

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7. Dashboard (Power BI)

An interactive dashboard was created using Power BI to visualize marketing performance.

The dashboard includes:

- **KPI cards for impressions, clicks, conversions, revenue, and cost**
- **Revenue by marketing channel**
- **Conversion distribution by channel**
- **Conversion rate trend over time**
- **Campaign efficiency scatter plot**
- **Top campaigns based on ROI**

The dashboard allows users to quickly identify which campaigns and channels perform best.

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8. Key Insights

From the analysis, several insights were discovered:

- **Some marketing channels generate significantly higher revenue than others.**
- **A small number of campaigns produce the highest ROI.**
- **Conversion rates vary across channels, indicating differences in audience engagement.**
- **Marketing performance changes over time, highlighting the importance of monitoring trends.**

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9. Business Recommendations

Based on the analysis, the following recommendations can be made:

1. Invest More in High-Performing Channels
Channels that consistently generate higher revenue and ROI should receive more marketing budget.

2. Optimize Low-Conversion Campaigns
Campaigns with many clicks but low conversions should be improved through better targeting and messaging.

3. Scale Successful Campaigns
Campaigns with high ROI should be analyzed and scaled to maximize marketing efficiency.

4. Monitor Campaign Performance Regularly
Tracking marketing performance over time can help businesses identify trends and adjust strategies.

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10. Conclusion

This project demonstrates how marketing campaign data can be analyzed using Python, SQL, and Power BI to generate meaningful insights.

By combining data cleaning, SQL analysis, and dashboard visualization, businesses can better understand campaign performance and make informed marketing decisions.